Underwater Vision-Based Surveillance Using Machine Learning and Artificial Intelligence

Day: 1 – Problem Understanding & System Definition

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1. Problem Understanding

Underwater vision-based surveillance involves analysing underwater images or video to obtain useful information about objects present in the underwater environment.

The system is expected to work with underwater visual data and associated object annotations. The main computer-vision tasks considered in the project are image enhancement, object detection, and, where sequential/video data is available, object tracking and stability analysis.

Underwater images may have visual-quality issues that can make objects difficult to identify. Therefore, image enhancement may be used to improve the visual quality of the input images before further analysis.

The object-detection stage is expected to identify relevant objects and determine their locations using bounding boxes, object classes, and confidence scores.

The detection results can then be evaluated against the ground-truth annotations using suitable evaluation measures such as precision, recall, and mAP.

If sequential/video data is available, detections across multiple frames can additionally be used for object tracking and stability analysis.

The exact experimental relationship between image enhancement and object detection, and the use of the available datasets, requires confirmation before implementation.

2. System Inputs

2.1 Underwater Images

Underwater images containing objects of interest.

2.2 Video / Sequential Data

Video or sequential frames, if such data is available for the project.

2.3 Object Labels

The supplied datasets contain annotation files corresponding to the images.

2.4 Object Annotations

The annotations provide information required for object detection, including:

1) Class ID
2) X-center
3) Y-center
4) Width
5) Height

2.5 Dataset Configuration

The data.yaml files provide information such as:

6) dataset paths
7) number of classes
8) class names

3. Dataset Understanding

Three datasets have been provided:

Dataset	Classes	Number of Classes
well.v8i.yolov8	Inlet-pipe, fishes, school-of-fish, stone	4
DIATAquarium.v4i.yolov8	Aquatic Plant, Camera, Conch-Shell, Fish, Fishes, Pluco, Shark, Temperature Sensor, Water-Pump, Water-filter, betta	11
Aquatic Plant.v2i.yolov8	Aquatic_plant	1

The datasets contain folders for: train, valid, and test — and each split contains images and labels subfolders. The images directory contains the actual underwater images, while the labels directory contains the corresponding object-detection annotations.

4. Understanding of YOLO Annotations

The supplied data uses a YOLO-style object-detection annotation format. Each annotation line follows the format:

class_id x_center y_center width height

For example:

1 0.0296875 0.08203125 0.059375 0.0671875

Here: 1 = class ID, 0.0296875 = bounding-box centre X, 0.08203125 = bounding-box centre Y, 0.059375 = bounding-box width, 0.0671875 = bounding-box height. The coordinates are normalized.

One image can contain multiple objects. Therefore, a single label file can contain multiple annotation lines.

5. Dataset Splits

The supplied datasets contain training, validation and test portions.

Split	Purpose
Training	Used to train the object-detection model
Validation	Used during model development and evaluation
Test	Used for final evaluation

During initial inspection, separate valid folders were observed in the datasets. However, the current data.yaml files for the Well and DIAT Aquarium datasets appear to point the validation path to the test/images directory even though separate valid/images directories exist. This configuration should be clarified before model experimentation rather than changed without confirmation.

6. Expected Outputs

6.1 Enhanced Image

An underwater image with improved visual quality, if image enhancement is included in the final pipeline.

6.2 Object Detection

For detected objects, the system should provide: object class, bounding box, and confidence score. For example: Fish — bounding box — confidence = 0.91.

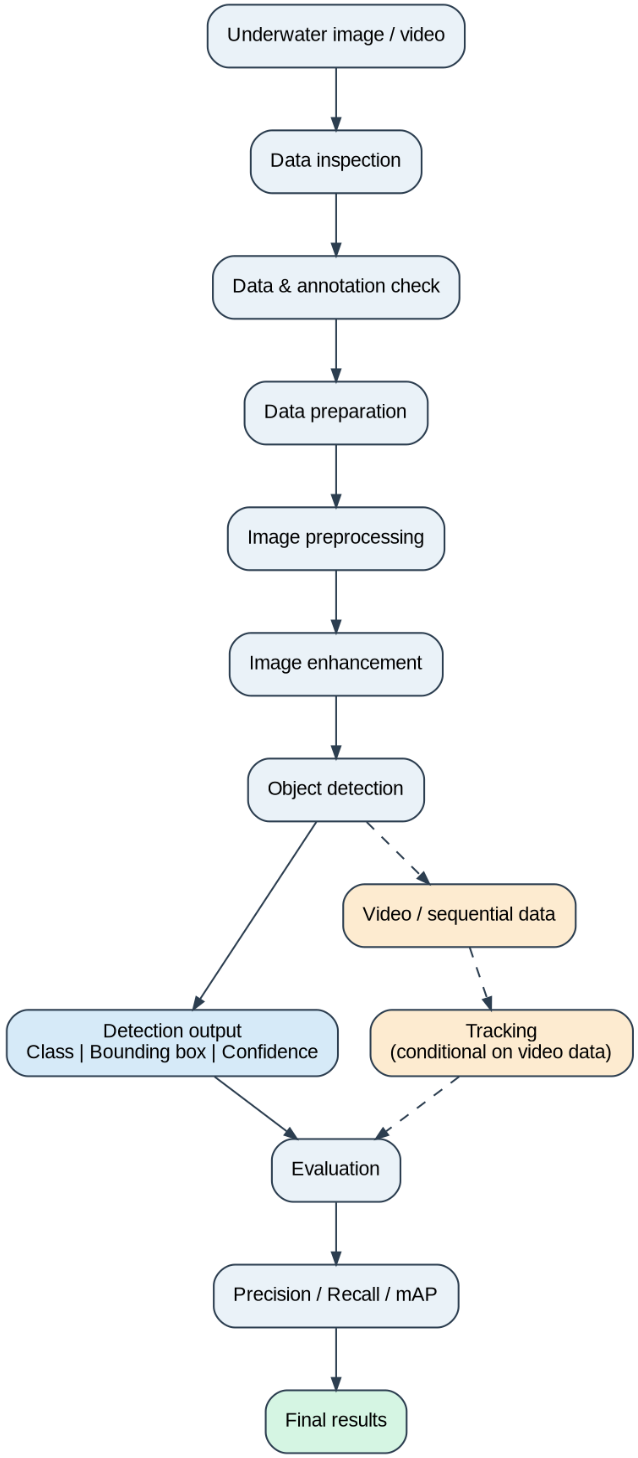
6.3 Detection Evaluation

The object-detection performance can be evaluated using measures such as precision, recall, and mAP. These metrics should be calculated after model predictions are compared with the ground-truth annotations.

6.4 Tracking / Stability Analysis

If video or sequential data is available, detected objects can be tracked across frames and the stability/consistency of the detections can be analysed.

7. End-to-End Software Pipeline



8. Success Criteria

The system will be considered successful if:

1) The supplied datasets can be correctly inspected and prepared.
2) Images and their corresponding annotations are correctly identified.
3) Dataset-quality issues can be identified before model training.
4) The image-preprocessing/enhancement stage can process the underwater images consistently.
5) The object-detection stage can identify relevant objects and their locations using bounding boxes.
6) Object-detection performance can be evaluated using appropriate metrics such as precision, recall and mAP.
7) If sequential/video data is available, object tracking and detection stability can be analysed.
8) The complete processing pipeline can be reproduced from input data to final results.

Note: no fake precision/recall/mAP numbers are included here — those are results to be obtained later after experiments.

9. Technical Questions / Assumptions Requiring Clarification

Q1. Should the three supplied datasets (well.v8i.yolov8, DIATAquarium.v4i.yolov8, and Aquatic Plant.v2i.yolov8) be used independently or combined?

Q2. Which dataset should be considered the **primary dataset** for the project?

Q3. Should the classes from the different datasets be kept as provided, or should similar classes be combined into a common class structure??

Q4. Should the underwater images be **enhanced first and then passed to the object-detection model**, or should image enhancement and object detection be evaluated separately?

Q5. Should object-detection performance on **original and enhanced images** be compared using precision, recall and mAP?

Q6. Is **video/sequential data** available for the project so that object tracking and stability analysis can be performed?

Q7. Which **object-detection model** and **image-enhancement method** should be used as the baseline?

Q8. Which **evaluation metrics** are mandatory for the final project, such as precision, recall, mAP, image-quality measures, or tracking metrics?

10. Current Understanding and Open Issues

1) The project involves **underwater visual data, object annotations, image enhancement, and object detection**.
2) **Object tracking and stability analysis** may also be required if sequential/video data is available.
3) Three **YOLOv8-format datasets** have been provided with different object classes.
4) The datasets contain **images and corresponding annotations** organized into training, validation, and test portions.
5) The exact strategy for **using the three datasets** is not yet confirmed.
6) The exact relationship between **image enhancement and object detection** is also not yet confirmed.